GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Autonomous Cloud Architect

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TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript (ES6+), Python, Go, SQL, HTML5, CSS3

Frontend & Spatial Web: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, Framer Motion, Lenis Smooth Scroll, Three.js, WebGL2 (Camera Kit), Core Web Vitals (LCP, CLS < 0.02, FCP)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB, Redis & BullMQ, WebSockets, REST APIs, Docker, Linux/Bash

Cloud, AI & Infrastructure: GitHub Actions 24/7 CI/CD Fleets, Meta Cloud API (WABA), Gemini AI Multimodal, FFmpeg, Railway, AWS (S3, Lambda), Exotel Telephony, Silero VAD, Sarvam AI, Groq LPU

WORK EXPERIENCE

CP Systems • Software Developer **2024 – 2025**
Software Engineering & Distributed Systems • Remote *Punjab, India*

- **Full-Stack Development & Internal Tooling:** Engineered web applications and backend services using TypeScript, Node.js, and SQL, improving data throughput and internal workflow automation.
- **Code Quality & Production Reliability:** Implemented automated CI testing pipelines, optimized database query execution plans, and collaborated across sprint cycles to deploy production features with 99.9% uptime.

Gurdharam Enterprise • Full-Stack Software Engineer & Solutions Architect **2025 – Present**
Independent Enterprise Software Consultancy • Remote *Punjab, India*

- **Commercial Client Delivery & High-Craft Web:** Architected bespoke web platforms, agency portals (Branders), and interactive 3D WebGL interfaces for clients in India and abroad, delivering sub-second load times and zero layout shift.
- **Telephony Voice AI & WhatsApp Infrastructure:** Built real-time telephony agent over Exotel WebSockets, Sarvam AI Indic LLMs, and Silero VAD (<700ms latency), alongside multi-tenant Meta Cloud API gateways (WABA ID: 106690...).

KEY PRODUCTION PLATFORMS & SYSTEMS

SOLUM Minerals — Procedural 3D WebGL Bio-Carbon Platform **2026 (Oct) – Present**
Lead Frontend Architect • Next.js 15, Three.js, WebGL2, Canvas API, Tailwind CSS Open Source

- **Procedural 3D WebGL Rendering:** Engineered code-generated Three.js WebGL tablet model synthesizing 1024x1024 noise textures procedurally in memory via Canvas API, eliminating 15MB+ of 3D asset downloads.
- **Scroll Kinematics & Core Web Vitals:** Implemented scroll-linked camera interpolation in Next.js 15 App Router, achieving a 99+ Lighthouse score with zero layout shift (CLS < 0.02) and sub-second FCP.

yt-auto — 24/7 Autonomous Cloud Media Fleet & Multimodal AI **2025 – Present**
Lead Systems Architect • Python, FFmpeg, GitHub Actions CI/CD Matrix, Gemini AI Open Source

- **Autonomous Cloud Fleet Infrastructure:** Architected 24/7 automated media production fleet operating across 5 distinct channels on GitHub Actions multi-worker matrices with zero local server dependencies.
- **Multimodal AI Judge & Dynamic Audio Pipeline:** Built dual-gate quality control pipeline with Gemini Multimodal Judge AI (≥85/100 threshold) and automated FFmpeg video composition with dynamic audio ducking.

EduCore — Enterprise College Management ERP **2025 – Present**
Lead Full-Stack Architect • React 19, TypeScript, Express, PostgreSQL, MariaDB, Railway Open Source

- **Dual-Engine Relational Architecture:** Engineered multi-tenant campus ERP supporting PostgreSQL and MariaDB connection pools with automatic connection health recovery, parameterized query sanitization, and sub-15ms latency.
- **9-Role RBAC & Dynamic Schema Builder:** Built strict JWT role-based access control across 9 permission tiers, paired with a runtime form schema builder and forensic audit trail logging actor IP, device profile, and data diffs.

Dream Heights — Luxury Spatial Real Estate Portal **2025 – Present**
Spatial & Frontend Architect • React 18, TypeScript, Tailwind CSS, Framer Motion, Lenis Open Source

- **Spatial Architectural Pins & Navigation:** Built luxury residence exploration portal in React 18 and Vite 6 featuring spatial residence pins, dynamic floor-plan modals, and custom interactive architectural emblems.
- **Hardware-Accelerated Physics & Inertial Momentum:** Integrated Lenis 1.1 smooth scrolling with Framer Motion layout transitions, maintaining silky 60 FPS rendering across high-DPI viewports.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) — B.Sc. in Computer Science **Expected May 2027**
Core: Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744) • Additional Projects: *SnapAR Studio (Camera Kit WebGL2)*, *TakeMyInterview AI*, *Fasal Doctor (Edge TFLite)*.